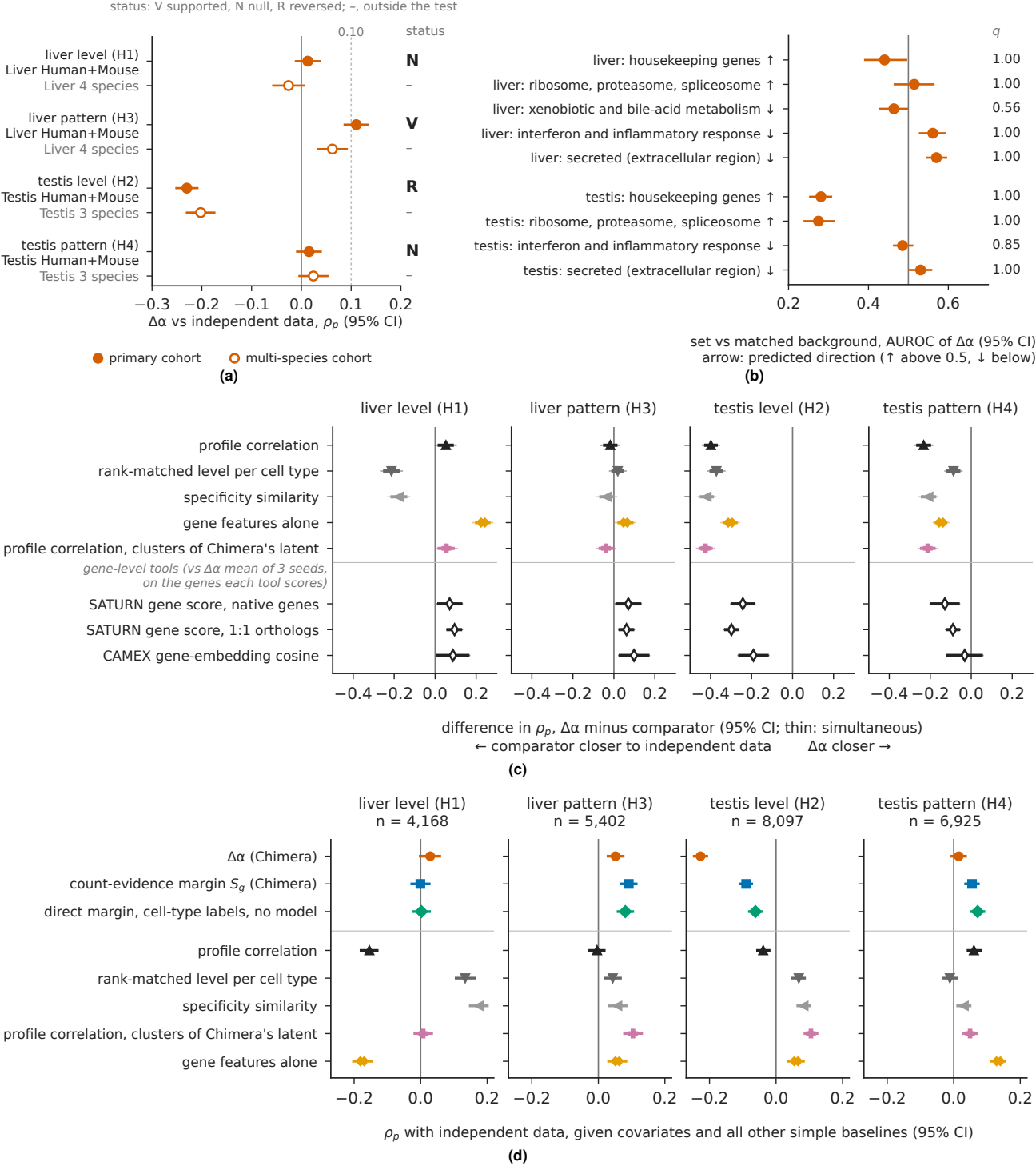

status: V supported, N null, R reversed; –, outside the test
0.10
status
liver level (H1)
Liver Human+Mouse
Liver 4 species
liver pattern (H3)
Liver Human+Mouse
Liver 4 species
testis level (H2)
Testis Human+Mouse
Testis 3 species
testis pattern (H4)
Testis Human+Mouse
Testis 3 species
N
–
V
–
R
–
N
–
−0.3 −0.2 −0.1 0.0 0.1 0.2
Δα vs independent data, ρ_p (95% CI)
primary cohort
multi-species cohort
(a)
q
liver: housekeeping genes ↑
liver: ribosome, proteasome, spliceosome ↑
liver: xenobiotic and bile-acid metabolism ↓
liver: interferon and inflammatory response ↓
liver: secreted (extracellular region) ↓
testis: housekeeping genes ↑
testis: ribosome, proteasome, spliceosome ↑
testis: interferon and inflammatory response ↓
testis: secreted (extracellular region) ↓
1.00
1.00
0.56
1.00
1.00
1.00
1.00
0.85
1.00
0.2 0.4 0.6
set vs matched background, AUROC of Δα (95% CI)
arrow: predicted direction (↑ above 0.5, ↓ below)
(b)
liver level (H1)
liver pattern (H3)
testis level (H2)
testis pattern (H4)
profile correlation
rank-matched level per cell type
specificity similarity
gene features alone
profile correlation, clusters of Chimera's latent
gene-level tools (vs Δα mean of 3 seeds, on the genes each tool scores)
SATURN gene score, native genes
SATURN gene score, 1:1 orthologs
CAMEX gene-embedding cosine
−0.4 −0.2 0.0 0.2
difference in ρ_p, Δα minus comparator (95% CI; thin: simultaneous)
← comparator closer to independent data
Δα closer →
(c)
liver level (H1)
n = 4,168
liver pattern (H3)
n = 5,402
testis level (H2)
n = 8,097
testis pattern (H4)
n = 6,925
Δα (Chimera)
count-evidence margin S_g (Chimera)
direct margin, cell-type labels, no model
profile correlation
rank-matched level per cell type
specificity similarity
profile correlation, clusters of Chimera's latent
gene features alone
−0.2 0.0 0.2
ρ_p with independent data, given covariates and all other simple baselines (95% CI)
(d)